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Nonlinear and threshold effects of the built environment, road vehicles and air pollution on urban vitality

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Abstract: The impact of factors such as the built environment, road vehicles, and air quality on urban vitality attracts increasing interest in urban planning and design research. However, tacit assumptions of linear relationships between these factors have been embedded in most studies, potentially leading to biased estimations of their effects on urban vitality. This study addresses the gap by using machine learning models and SHAP (SHapley Additive exPlanations) to investigate the non-linear relationships and threshold effects between urban vitality, the built environment, road vehicles, and air pollution in Manhattan, NYC. Urban vitality was represented by pedestrian presence in 29,540 street-view images. Results showed that Extreme Gradient Boosting outperformed Ordinary Least Squares, Random Forest, and Gradient Boosting Decision Trees in urban vitality estimation. It reveals that while built environment variables accounted for 77.5% of the explanation for urban vitality, road vehicles (such as bicycles, buses, cars and motorbikes) and ozone concentrations accounted for 15.18% and 1.46%, respectively. The built environment and road vehicle factors exhibit positive nonlinear relationships with urban vitality. Meanwhile, ozone concentration demonstrated a negative threshold effect on urban vitality with a threshold at 27.5 ppb. This study advances our understanding of the threshold effect mechanism of the built environment, road vehicles, and air pollution on urban vitality, offering practical recommendations for improving urban vitality.

Keywords: *urban vitality, street view image, non-linear relationship, air pollution, road vehicles*

1. Introduction

Urban vitality is a crucial aspect of sustainable urban development, as it reflects the attractiveness of areas in terms of human and activity dimensions (March et al., 2012). Numerous studies have demonstrated that public spaces and human activity are the driving forces behind urban vitality (Jalaladdini and Oktay, 2012; Xia et al., 2020). Urban vitality is essential to the liveliness of public spaces because it is closely linked to human interactions and pedestrian activities (Ouyang et al., 2022), which are necessary to attract talent, capital, and businesses (Chen et al., 2022c). Areas with high urban vitality offer a wealth of employment opportunities and improved quality of life (Liu et al., 2022). Urban planners have become increasingly interested in understanding how to design cities that promote liveliness and vibrancy. Understanding the intrinsic mechanisms of urban vitality and its determinants can help promote vitality in urban spaces, ultimately contributing to the overall sustainability of urban development.

Many previous studies have examined the associations between urban vitality and the built environment (BE) (Mouratidis and Poortinga, 2020; Xia et al., 2020). Li et al. (2022b) found that neighbourhood characteristics, urban form, function, landscape, location, and street configuration all contribute to enhancing urban vitality, with recreational and transportation services playing a vital role. Mouratidis and Poortinga (2020) uncovered the association between urban vitality, 3D built environment, and social cohesion, emphasizing the significance of urban amenity density and proximity to the city centre. Chen et al. (2023) explored the relationship between the BE features and urban vitality using Ordinary Least Square (OLS), Geographically Weighted Regression (GWR) and mobile phone data, revealing the importance of floor area ratio and road density. Similarly, Wang et al. (2022a) investigated the impact of point of interest

(POI), socioeconomic status, and location accessibility on urban vitality using the spatial lag model (SLM) and Multiscale Geographically Weighted Regression (MGWR). Li et al. (2022c) examined the relationship between street width, commercial density and urban vitality using a linear regression model, finding that street width has a considerable positive influence on street vitality. However, these studies often relied on linear assumptions (mostly OLS and GWR) to determine the relationship between urban vitality and the built environment, which may not always accurately reflect the complex interplay between these factors. A more nuanced approach may be necessary to better understand the relationship between urban vitality and the built environment, accounting for the varying effects of different factors and their interactions.

Recent research in urban planning has increasingly recognized the non-linear and threshold effects of the built environment (BE) features on human behaviors (Zhang et al., 2020b) and social issues (Kim and Lee, 2023). Neglecting these non-linear associations in urban vitality studies may result in inaccurate urban planning implications and policies. As a result, there is a growing emphasis on understanding the non-linear relationship between urban vitality and the BE. For example, studies by Yang et al. (2021), Xiao et al. (2021), and Peng et al. (2023) have explored the non-linear effects of various aspects of the built environment, such as land use, subway accessibility, and other built environment features on urban vitality. While these studies have provided valuable insights into the non-linear mechanisms between the urban environment and urban vitality, they have some limitations. First, data used to measure urban vitality often comes from private sources like mobile phone companies, which may not fully represent pedestrian activity on the street. Second, previous studies have focused on single dimensions or limited aspects of the built environment, overlooking the complex and multifaceted nature of the

built environment's impact on urban vitality. Third, spatial analysis has been mainly conducted at neighbourhood or metro station levels, leaving micro-spatial analysis underexplored.

Urban vitality is closely related to street activities (Jalaladdini and Oktay, 2012), which are influenced by various factors, including transportation, pollution, and the built environment. Transportation options like bicycles and buses promote physical activity and foster a healthy lifestyle (Patterson et al., 2019). Studies have shown that initiatives to increase public transport usage are associated with higher levels of physical activity and social engagement (Lavery et al., 2018). Despite their significance, the impact of transportation modes on urban vitality has yet to be extensively studied. The rising concern over air pollution in many cities underscores its potential to affect urban vitality negatively. Pollution can pose significant health risks, prompting individuals to adjust their activities to mitigate exposure (Huang et al., 2023). For instance, people may reduce outdoor activities such as dining out, shopping, and walking to minimize pollution exposure (Cheng et al., 2020). Consequently, pollution not only compromises public health but also dampens the vibrancy of urban spaces. Furthermore, the interactions between transportation modes, pollution, and the built environment can synergistically affect urban vitality. However, empirical evidence of these complex, nonlinear relationships remains limited. Understanding the intricate interplay between these factors is crucial for developing effective strategies to enhance urban vitality and promote sustainable urban development.

Aggregating human movement in public urban spaces is commonly used as a proxy for assessing urban vitality (Montgomery, 1998). Traditionally, pedestrian volume has been assessed through field surveys to characterize street vitality (Sung and Lee, 2015). While insightful, this approach can be costly and impractical on a large scale. Researchers have employed alternative metrics, including nighttime light data (Lan et al., 2020), the built environment (Paköz et al.,

2022), and catering services (Xia et al., 2020), to estimate urban vitality. However, these metrics often provide static snapshots of the urban environment. With the advent of mobile phones and social media data, scholars have turned to these sources as proxies for urban vitality. While these data sources offer valuable insights, they also present limitations, such as privacy concerns and incomplete representations of urban activities. Alternatively, street view images (SVI) combined with machine learning techniques have emerged as a promising tool for studying urban dynamics (Ye et al., 2019). For instance, this approach has been used to evaluate urban greenness (Ki et al., 2023), public facilities (Kelly et al., 2013), and the urban visual environment (Liang et al., 2023). Recent studies have also demonstrated the effectiveness of SVI in estimating pedestrian volume, a critical indicator of urban vitality (Yin et al., 2015; Chen et al., 2022a). Additionally, machine learning techniques and interpretation models have been acknowledged for their ability to explore complex relationships, such as non-linear and threshold effects, in urban studies (Liu et al., 2023).

To fill these gaps, this study aims to quantitatively examine the non-linear relationship between the built environment, road vehicles, air pollution and urban vitality using street view images and machine learning models. First, pedestrians in street view images are counted using an object detection technique to indicate urban vitality. Second, we integrated the built environment, pollution and road vehicle volume into a 100x100m grid for micro-level analysis as Delclòs-Alió et al. (2019). Machine learning and SHapley Additive exPlanations (SHAP) are then employed to analyze and interpret the results. To the best of our knowledge, this is the first research to explore the non-linear impact of built environment, vehicle presence, air pollution and urban vitality. The contribution of this work includes (1) Uncovering the relative importance of the built environment features, vehicles, and pollution on urban vitality; (2) Explaining the

non-linear and threshold effects of explanatory variables on urban vitality; (3) Offering insights for urban planners and policymakers to promote urban vitality.

The remainder of this work is structured as follows. Section 2 provides a comprehensive overview of urban vitality measurement and the potential non-linear relationship between urban vitality and the BE, road vehicles and pollution. Section 3 presents the study area, data, and methodology. Section 4 describes the main results, including relative importance and non-linear relationship between variables. Finally, Section 5 and Section 6 discuss the findings and conclude, respectively.

2. Literature review

2.1. The measurement of urban vitality

Urban vitality, a term introduced by Jacobs (1961), has undergone significant evolution over the decades. Initially, it referred to the dynamic and diverse urban life in residential areas, encompassing various social activities. However, as urban studies progressed, urban vitality's definition expanded to preserve internal growth and development resulting from a well-designed and desirable urban environment (Lynch, 1984). According to Maas (1984), urban vitality is influenced by several factors, including socially diverse pedestrians and a wide range of unique commercial and leisure options, all contributing to a city's spatial quality and vibrancy. Meanwhile, Montgomery (1998) further emphasized the significance of urban vitality by defining it as the degree to which an area feels alive or vibrant. He highlighted the close relationship between the vitality of a city's streets and its overall urban activity. The value of increasing population density in public spaces has been recently highlighted (Shach-Pinsly, 2019), as it enhances one's sense of personal security and fosters urban vitality through increased human activities.

As interest in urban vitality grows, researchers have employed various indicators to estimate this dynamic concept. Approaches such as nighttime light data (Gong et al., 2022), the built environment (Yue et al., 2021) and land use diversity have been used to gauge urban vitality. For instance, Delclòs-Alió and Miralles-Guasch (2018) and Wu et al. (2022b) recreated urban vitality conditions through diversity, accessibility, density and design dimensions. However, these indicators primarily reflect the static aspects of the urban environment, while vitality is inherently dynamic. In response, scholars have turned to urban big data and emerging technologies to capture the dynamic essence of urban vitality.

Researchers can access real-time insights into human behavior and urban dynamics using mobile phone data and social media sources. Studies have leveraged Baidu heat maps to measure walking activity and social check-ins on platforms like Facebook, Twitter, and Weibo to gauge social engagement. For example, Wu et al. (2018a) used check-in records on weekdays and weekends to measure urban vitality and its association with land use configurations. Chen et al. (2022c) and Xia et al. (2020) used small catering to represent urban vitality at the block level. Zhang et al. (2022) proposed using taxi drop-off points to estimate urban vitality. Additionally, scholars have leveraged GPS data from mobile phone users (Kim, 2020; Botta and Gutiérrez-Roig, 2021) as the primary indicator for measuring urban vitality. Smart cards, like metro and bank card transactions, were employed to reflect urban vitality (Kim, 2018). However, the data types used in the studies above have some limitations. While small catering and food facilities only reflect a small aspect of urban vitality, taxi drop-off data only reveal the start-end point of the journey and cannot reveal pedestrian activities. Moreover, mobile phone data, though valuable, can pose privacy concerns and may require appropriate permissions or ethical

considerations. Some scholars used socio-economic to compare the urban vitality of cities (Shi et al., 2021; Liu et al., 2022), but this approach faces challenges in micro-scale measurement.

Recently, scholars have further validated the measurement of urban vitality using street-view images by comparing pedestrian volume data derived from field surveys with street-view image data (Liu et al., 2018). This approach has been used in recent studies to measure urban vitality (Li et al., 2022b; Li et al., 2022c). Street view images have become an increasingly popular data source in urban research due to their ability to capture urban elements effectively (Kang et al., 2020). Street view images allow for detailed and extensive coverage, facilitating the extraction of large-scale streetscape attributes, such as pedestrian volumes, through modern image-processing techniques (Biljecki and Ito, 2021). Many studies have used street view images to identify urban vitality (Qi et al., 2020; Ma, 2023). Therefore, this study used pedestrian volume in street view images to represent urban vitality.

2.2. The relationships between the built environment, road vehicles, air pollution and urban vitality

Previous studies have primarily explored the relationship between urban vitality and the built environment with linear assumptions. They emphasized factors such as density, diversity, distance to transit, design, and destination accessibility in promoting urban vitality. The following are key points concerning the relationship between urban vitality, built environment (density, diversity, distance to transit, design, destination accessibility), road vehicles and air pollution.

Most studies have reported a positive linear association between density and urban vitality. For instance, Wu et al. (2022a) found a statistically significant relationship between building density and urban vitality, emphasizing the importance of urban concentration to urban vitality. Similarly, Fan et al. (2021) found that built density has the most significant impact on

urban vitality in waterfront areas, a finding also observed by Ye et al. (2018). However, it is worth noting that Li et al. (2022b) reported the negative impact of road density and urban vitality, while Lv et al. (2022) discovered a significantly positive association.

Regarding diversity, land use and POI mix are the most frequently used features in urban vitality studies. Mouratidis and Poortinga (2020) regarded land use diversity as an effective strategy to integrate physical functions and enhance social connectivity, ultimately promoting urban vitality, aligning with the ideas of Jacobs (1961). Xia et al. (2020) and Wangbao (2022) observed a positive association between land use diversity and urban vitality, noting that an increased land use mix degree contributes to enhanced street vitality. However, Liu and Shi (2022) suggested that studies should consider interaction effects between variables because they found that when accounting for factors such as floor area ratio or unintended impacts, the correlation between land use diversity and urban vitality might turn negative, which is consistent with the study of Xia et al. (2022).

In terms of distance to transit, the characteristics of transportation infrastructure are intricately linked to urban vitality. Proximity to transit and transportation facilities significantly influences urban vitality (Xia et al., 2022). Vibrant blocks are typically characterized by effective transport services and easy access to subway and bus stations (Zhang et al., 2020a). Notably, subway stations play a significant role in enhancing street vitality, while bus services contribute to the vibrancy around metro stations, as evidenced by the findings of Xiao et al. (2021). Using the OLS model and taking Seoul as a case study, Lu et al. (2019) revealed a positive correlation between public transport accessibility (bus, subway) and street vibrancy.

Design is considered a crucial feature which creates a vivid urban environment. Among design features, building type is considered more influential than density in shaping urban

vitality (Ye et al., 2018). Block and strip structures positively impact vitality, even when accounting for building density. Smaller block sizes are also believed to promote walking and interactions, thus contributing to urban vitality, as observed by Jacobs (1961) and Paköz et al. (2022). Sung and Lee (2015) found a negative effect between urban vitality and block size. Additionally, the age of buildings can impact urban vitality, with older buildings exhibiting a positive correlation with street vitality in high-density urban areas (Wangbao, 2022).

Destination accessibility plays a crucial role in fostering a lively city. Studies on the relationship between destination accessibility and urban vitality typically used Point of Interest (POI) accessibility, as shown in previous studies (Yue et al., 2017; Li et al., 2021a; Wangbao, 2022). The results indicated that destination accessibility is essential to fostering vibrant cities.

Table 1 presents an overview of the literature on related studies that significantly advance understanding of urban vitality, built environment and pollution. However, certain limitations within these studies need further investigation to comprehensively assess the effects of the urban environment on urban vitality. The first limitation is related to data sources, which have been limited, leading previous studies to mainly focus on analyzing pairwise relationships between urban vitality and specific built environment factors. However, it is essential to recognize that the built environment indicators may have interconnected relationships like what is observed daily. Not accounting for these interconnections might result in blurred effects of urban vitality and potentially lead to inflated or underestimated relationship estimations (Li et al., 2022b). Therefore, it is essential to fully explore various aspects of the built environment on urban vitality. The second significant limitation is using linear assumptions to explore the potential relationship between urban vitality and the built environment. This approach assumes that urban vitality will increase linearly as the values of the built environment indicators increase,

potentially leading to unrealistic estimations. Non-linear effects are common in the context of urban development. Thus, the effects of features may only be positive or negative within a certain range, exhibiting non-linear effects. Neglecting these non-linear effects could result in an incomplete understanding of the impact of the built environment and air pollution on urban vitality. Thirdly, vehicles, including bicycles, buses, and motorbikes, play a significant role in urban mobility. These vehicles supply essential mobility options to access various destinations within the city. It fosters a sense of connectivity, making these areas more attractive for residents and visitors alike. However, empirical evidence on the relationships between urban vitality and the presence of these modes of transport is far from enough. Finally, air pollution significantly affects human health, so people prefer less polluted areas that may influence urban vitality. However, very few studies considered the effects of environmental pollution on urban vitality. We only found a study by Lee and Kang (2022), who used the structural equation modelling (SEM) model to explore the negative impact of PM_{2.5} on urban vitality. However, the complex mechanism of air pollution on urban vitality has not been fully explored. To what extent air pollution declines urban vitality and the threshold effect are questions needing further investigation.

Table 1. Literature on the built environment and vitality

Authors	Type of relationship	Representative for urban vitality	Method
(Sung and Lee, 2015)	The built environment and walking activity	Walking time	OLS
(Wu et al., 2018b)	Neighbourhood vitality and urban form	GPS from 534 participant	OLS
(Zhang et al., 2020a)	Urban landscape and urban vitality	Small catering and nighttime light	OLS, GWR
(Meng and Xing, 2019)	Urban landscape characteristics and urban vitality	Comments of users on a social platform	OLS
(Lv et al., 2022)	The built environment and block vitality	Baidu heat map	GWR

Authors	Type of relationship	Representative for urban vitality	Method
(Li et al., 2021b)	Street network and urban vitality	Baidu heat map	SLM, SEM
(Li et al., 2022b)	The built environment and urban vitality	People and vehicles from street-view images	SLM
(Li et al., 2022c)	Street-built environment and urban vitality	Pedestrians from a video camera	OLS
(Wu et al., 2022c)	The built environment and street vitality	GPS mobile phone	Geary's C; SLM; SEM
(Wang et al., 2022b)	POI mix and urban vitality	Mobile phone data	Bayesian
(Gong et al., 2022)	The built environment and urban vitality	Nighttime light, Baidu heat map, culture POI	OLS, GWR
(Tu et al., 2022)	The built environment and metro vitality	Smart card	OLS, SLM
(Lee and Kang, 2022)	The built environment, particulate matter and urban vitality	Visiting population in panel data	SEM, SDM
(Xiao and Liu, 2023)	The built environment and urban vibrancy during Covid 19	Review volume per restaurant	NGBoost

3. Material and methodology

We developed an analytical framework to quantify the non-linear relationship between urban vitality, BE, road vehicles and air pollution. Figure 1 depicts the analytical framework, which includes data preparation and analysis. Firstly, Data preparation, which consists of street view image detection, air pollution, and GIS data collection, is carried out. During the detection stage, this study counts objects in each image, such as humans, cars, buses, motorbikes, and bicycles. Before data analysis, we integrated the data into a 100 x 100 m grid. Then, we used OLS and multicollinearity tests to select significant variables before employing non-linear machine learning models. Next, the relative importance of explanatory variables and non-linear and threshold identification were determined using the most accurate machine learning algorithms and SHAP. In the following subsection, data and methods are detailed.

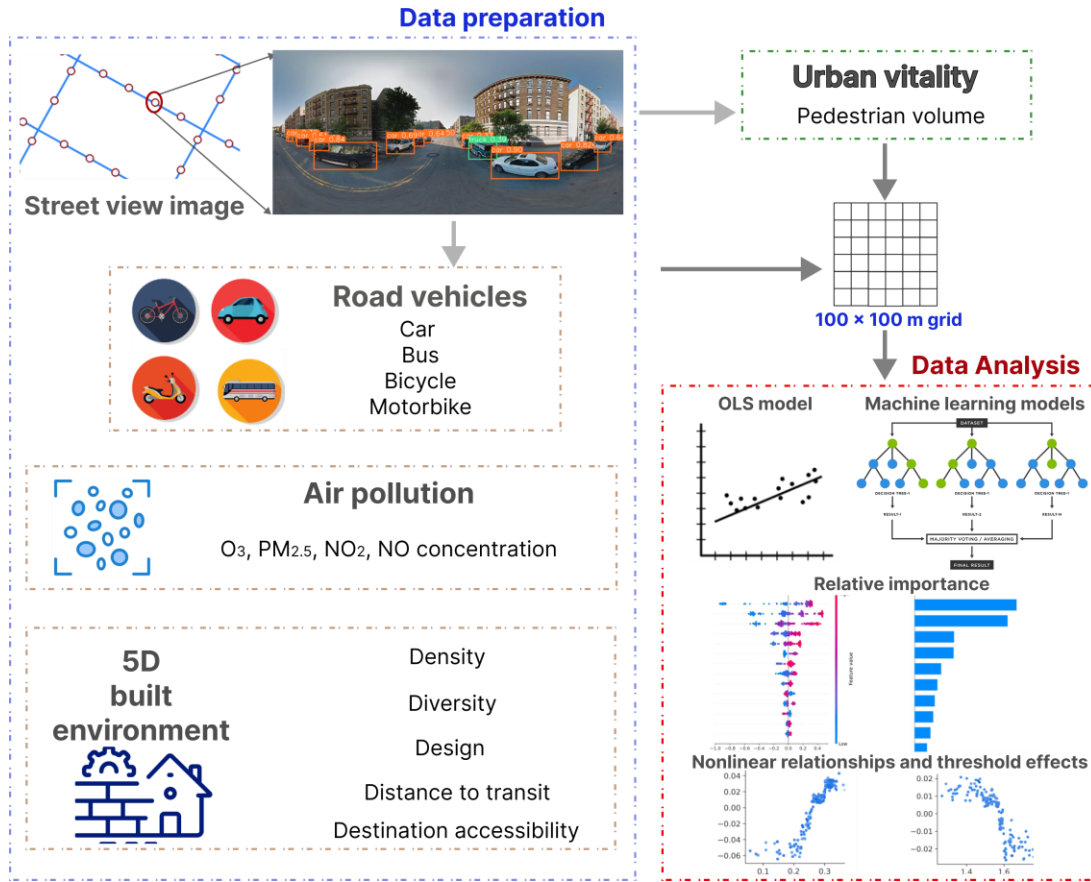


Figure 1. The analytical framework

3.1. Study area

Manhattan is a captivating and iconic urban center that epitomizes the bustling energy of New York City (NYC). Manhattan is situated at the heart of the metropolis, attracting millions of residents and visitors from around the globe with its vibrant and diverse character (figure 2). Manhattan has an estimated population of 1.63 million people over approximately 22.7 square miles. This densely populated city features a diverse blend of cultures, architecture, landmarks, and economic activities, all of which have made a lasting impression on the world. Manhattan is the most densely populated of the five NYC boroughs, with a population density of 27,346 people per square kilometre. Manhattan's average income is 89,008 USD per capita, nearly double the citywide average. However, the borough also grapples with a poverty rate of 17.3%,

approximately 25% higher than the citywide average. Manhattan accommodates 929,239 housing units, with 87% currently occupied. The number of households totals 803,844, with an average of 1.9 persons per household. Urban planners and policymakers face challenges related to sustainable growth, affordable housing, transportation infrastructure, and environmental conservation to ensure Manhattan's resilience and livability for future generations.

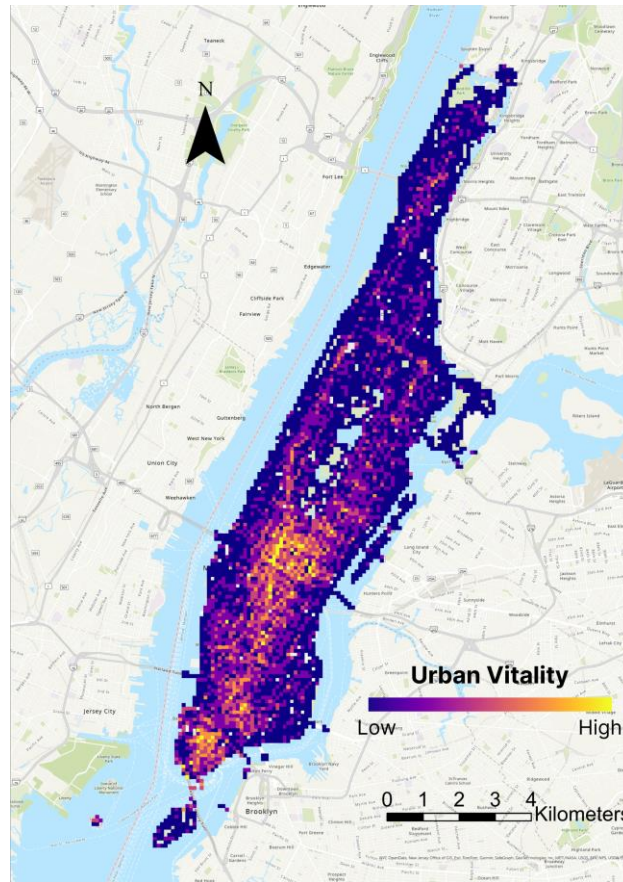


Figure 2. Location of Manhattan district, New York City

3.2. Data sources

Three main data types were employed in this study: street view images, air pollution, and the built environment data.

3.2.1. Street view images

Using real-world images in research has recently become a prominent trend, including street view images and social media photos (Biljecki and Ito, 2021). Street view images offer valuable visual features that provide insights into urban environments. In this research, we used street view image data retrieved through the Google API to approximate pedestrian activity, serving as an indicator of street vitality. Images were collected at 30-meter intervals along the street network, covering the Manhattan district (Figure 3). In total, 29,540 sample points were created, resulting in the collection of 82,180 panorama images and the generation of 29,540 mosaic images. The street view images were captured in four directions (0°, 90°, 180°, 270°). The images were taken between February 2021 and January 2022, encompassing the period affected by the COVID-19 pandemic. To account for the potential influence of COVID-19 on urban vitality, we divided the study period into two distinct phases: the COVID-19 outbreak period and the reopening period. The COVID-19 outbreak period spanned from February 2021 to May 2021 and November 2021 to January 2022, while the reopening period extended from June 2021 to October 2021. Considering the seasonal variation in pedestrian volume, we classified the collected images into four seasons: Spring (April-June), Summer (July-September), Autumn (October-December), and Winter (January-March). The geolocations of the images were derived from the names of each image file and are coded as “latitude-longtitude.jpg”.

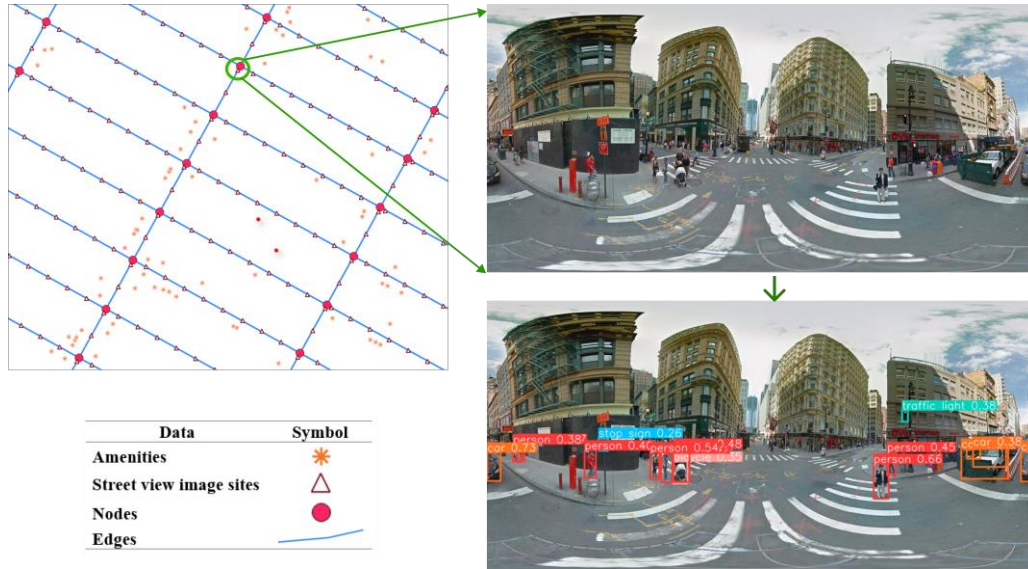


Figure 3. Example of data in a grid 100m x 100m¹ and object detection in an angle (90°)

3.2.2. The built environment and pollution data

The built environment features were selected based on the 5Ds built environment framework and literature review. The built environment data was extracted from the MapPLUTO database of 2021 provided by the Department of City Planning, NYC². 5Ds built environment stands for density, diversity, design, destination accessibility and distance to transit (Ewing and Cervero, 2010). The density was measured by calculating the built floor area ratio, residential floor area ratio, commercial floor area ratio, facility floor area ratio, number of buildings, floors, and residential units as in previous studies (see Appendix 2). For diversity, this study used the mixed point of interest (mixed POI), calculated from POI data downloaded from Overpass Turbo, a reliable granular source providing information on urban land use. POI was categorized into 11 groups, illustrating various urban functions (catering services, shopping services, residential services, financial services, life services, recreational services, corporation services, science & technological services and school, medical services, transportation services,

¹ The grid size is exaggerated for visualization purpose.

² <https://www.nyc.gov/site/planning/data-maps/open-data/dwn-pluto-mappluto.page>

government, and social organization) (see Appendix 1). As for design, we calculated the number of road nodes and edge length, the average lot's front, building depth, and commercial, retail, and garage areas in each grid. Destination accessibility was measured based on the number of POIs. However, as a POI in a cell can attract pedestrians from other cells, following previous studies, this study used the Kennerl density as a proxy for destination accessibility. The higher POI density means higher destination accessibility. The distance to transit was calculated as the density of bus stops, subway, and railway stops. Moreover, we also include the air pollution variables to explore how air pollution affects urban vitality, which is still a gap in the body of literature. The air pollution data includes the average predicted surface levels of ozone, PM_{2.5}, and NO₂ concentrations in 2021, with a resolution of 250m collected from NYC Open Data. The descriptive statistics are shown in Appendix 2.

3.2.3. Other data

Population density data were sourced from WorldPop³ with a resolution of 200 meters per pixel. WorldPop is widely recognized as a reliable data source and has been utilized in numerous urban studies. The housing unit data and income were obtained from the United States Census Bureau, providing detailed information at the census block level.

3.3. Methodology

3.3.1. Image processing and data aggregation

Before data aggregation, street view images were analyzed to detect pedestrians and vehicles. This process involved employing the Object Detection Model (ODM) - YOLOv5, an advanced object recognition approach. Building on the original YOLO series proposed by Redmon et al. (2016), YOLOv5 brings notable advancements to object detection. This unique

³ [Open Spatial Demographic Data and Research - WorldPop](#)

neural network-based model demonstrates exceptional performance and real-time inference capabilities, as demonstrated by studies conducted by Huang et al. (2018); Shorfuzzaman et al. (2021). Additionally, YOLOv5 exhibits exceptional model generalization, enabling it to detect various objects effectively, as shown by Jiang et al. (2022). Compared to its predecessors, YOLOv5 demonstrates superior detection accuracy, reduced mega floating-point operations, and smaller model sizes. Although YOLOv5 is not the latest version, its performance in object detection still achieves impressive results, employed by many recent studies (Li et al., 2022a; Inbar et al., 2023; Ren et al., 2023).

This study used the YOLOv5 trained on the COCO (Common Objects in Context) dataset to detect humans and vehicles (including bicycles, buses, cars, and motorcycles). The trained YOLO model was then deployed to perform real-time inference on the street view images, detecting human objects with bounding boxes encapsulating their spatial extent. To assess the accuracy of the detection results, 200 randomly selected images underwent manual auditing. The comparison between YOLOv5 detection and manual audit yielded high consistency, as evidenced by Pearson's correlation coefficients. Specifically, the correlation coefficient for human detection was 0.89, indicating a strong positive correlation between YOLOv5 and manual audit. Similarly, for car, bus, bicycle, and motorcycle detection, Pearson's correlation coefficients were 0.92, 0.90, 0.90, and 0.93, respectively, indicating a high level of consistency between YOLOv5 and manual audit results.

A spatial unit of analysis was defined to analyze the relationships between street vitality and built environment, road vehicles and air pollution characteristics. Regular square cells ($D_{side} = 100$ meters) were created for analysis, covering the entire study area, resulting in 5679 squares. Data collected within each square is aggregated to each grid. For example, edges are combined

by calculating the total length of edges within the cell. Meanwhile, the number of nodes that fall within each cell is counted to aggregate nodes. The person class is aggregated by dividing the total number of detected individuals in all images of a square by the number of processed SVI within the same square. POI density was calculated using the Kernel density function and then assigned to the respective cell. Moreover, to mitigate the impact of dimension and accelerate model training, we applied z-score normalization for some numerical variables. The z-score normalization is calculated as follows: $z\text{-score} = \frac{x-\mu}{\sigma}$; where μ is the mean value, σ is the standard deviation.

Previous studies show that the POI mix is vital in promoting urban vitality. Therefore, this study calculated the POI mix value for each cell. The calculation of the POI mix is calculated using Entropy index shown as follows:

$$\text{POI mix} = \sum_i^t \frac{W_i \cdot \ln(W_i)}{\ln(t)} \quad (\text{Eq.1})$$

Where W_i represents the proportion of i^{th} POI category, t represents the number of POI categories.

3.3.2. XGBoost

This study employed four prediction models: Ordinary Least Squares (OLS), Gradient Boosted Decision Trees (GBDT), Random Forest (RF) and eXtreme Gradient Boosting (XGBoost). The findings indicated that XGBoost demonstrated superior performance compared to OLS, RF and GBDT. XGBoost is a powerful non-linear machine learning technique that employs boosting to create highly accurate predictive models. Introduced by Chen and Guestrin (2016), it has gained popularity across various domains for its exceptional performance. The core algorithm of XGBoost is gradient boosting, an ensemble method that combines weak learners to

form a more powerful learner. Each subsequent weak learner addresses its predecessor's limitations, improving predictive capabilities. XGBoost is renowned for its versatility and effectiveness in classification and regression tasks.

In machine learning models, selecting appropriate parameters is crucial for achieving optimal performance. For XGBoost, key parameters such as `n_estimators`, learning rate, max depth, `min_child_weight`, `subsample`, and `gamma` play significant roles in determining model performance. The number of boosting rounds or trees (`n_estimators`) influences model complexity and computational time, with higher values potentially leading to better performance at the risk of overfitting. The learning rate controls the step size during each boosting round, with lower values promoting slower learning but improving generalization and preventing overfitting. Through optimization, we found that setting `n_estimators=500`, `learning_rate=0.01`, `subsample=0.8`, `gamma=0.1`, `colsample_bytree=0.9`, and `min_child_weight=3` yielded the highest R-squared value of 0.560 for XGBoost, indicating better model performance. These parameters were chosen based on their ability to balance model complexity and generalization. Similar optimization procedures were applied to the other two models.

3.3.3. SHAPley

Existing machine learning techniques are black boxes as they cannot interpret the system's structure. Lundberg and Lee (2017) proposed SHAP based on game theory to machine learning interpretation. The SHAP approach offers a comprehensive analysis of the global, local, and interactive effects, which cannot be effectively captured by partial dependence plots and accumulated local effect methods. In this model, the Shapley value is utilized to determine the contribution level of explanatory variables to the prediction variable. Furthermore, SHAP is model agnostic, meaning it can be applied to any machine learning model regardless of its

architecture or underlying algorithm. This flexibility allows SHAP to be used with various models, including RF, GBDT, XGBoost, neural networks, and support vector machines. Combining machine learning techniques with the SHAP method can uncover the relative importance of variables, non-linear relationships, thresholds, and variable interactions. In this study, we utilized variable importance to rank the variables in terms of their significance. Following previous research, SHAP values were employed to elucidate the non-linear impacts of the BE, vehicles, and air pollution on urban vitality (Kim and Lee, 2023; Doan et al., 2024).

4. Result

4.1. Exploratory variable selection

Following previous studies (Doan, 2023), this study employed the OLS model to select significant BE, road vehicles and air pollution variables that potentially influence urban vitality using Variance Inflation Factor (VIF) value and statistical significance. The selected variables are at 5% significant confidence and $VIF < 7.5$. Among 47 variables, 36 variables remained in our model. The VIFs of dependent variables are below 7.5, indicating that the model does not exhibit multicollinearity.

The OLS result is presented in Table 2. Regarding vehicles on the street, bicycles, motorcycles, and buses are positively associated with urban vitality, with the highest coefficient = 1.592 for bicycles. The appearance of cars also positively impacts urban vitality despite the negligible effect. As expected, pollution concentration negatively influences urban vitality, with the coefficient = -0.41, $p\text{-value} = 0.000$. This initial result confirms that air pollution might decline urban vitality.

Regarding the design, the number of nodes, buildings, and building depth is positively associated with urban vitality. However, the number of floors and lot frontage exhibit negative influences on urban vitality; that is, an increase in lot frontage leads to a decrease in urban

vitality. This finding aligns with the studies of Xia et al. (2020) and Yue et al. (2021), who pointed out the importance of small blocks in promoting urban vitality.

Density is also an important dimension significantly associated with urban vitality. While built floor area ratio, commercial floor area ratio and facility floor area ratio positively impact urban vitality, residential floor area ratio is negatively associated. Noticeably, destination accessibility exhibits significant impacts on urban vitality. Catering, residential, financial, science, and medical services positively influence urban vitality. While financial services have the highest impact on urban vitality, government services exhibit the lowest influence.

As for distance to the transit, transportation services significantly impact urban vitality, with a coefficient of 0.877. Finally, diversity significantly contributes to urban vitality, with a coefficient of 5.71 for the mixed POI.

Aligned with previous studies, we found that urban vitality varies by season. Street activities are most vibrant during the summer and autumn, followed by spring while achieving the lowest levels during winter due to low temperatures and the COVID-19 outbreak in Manhattan. Our findings are consistent with studies by Aultman-Hall et al. (2009); Miranda-Moreno and Lahti (2013), who observed the sensitivity of pedestrians to wind speed and cold temperatures. Additionally, we observed the negative impact of COVID-19 on urban vitality, which aligns with the findings of Delclòs-Alió et al. (2023), who noted a significant decrease in the walking frequency and time spent on daily walking during the COVID-19 period, contributing to reduced urban vitality (Xiao and Liu, 2023).

Overall, the OLS model revealed initial patterns in the relationship between urban vitality, built environment, road vehicles and air pollution. However, the OLS cannot identify the threshold effects to clarify to what extent independent variables influence urban vitality. When

do the explanatory variables begin to influence urban vitality, and are their effects consistent across different ranges? In the following step, non-linear machine learning models and an explainable technique- SHAP- will answer these concerns.

Table 2. OLS model performance

Term		Estimate (std)	t
Vehicle	(Intercept)	15.9914*** (1.011)	15.812
	Bicycle	1.5929*** (0.146)	10.904
	Car	0.0160*** (0.003)	4.573
	Motorcycle	0.6616*** (0.135)	4.895
	Bus	0.2582*** (0.046)	5.615
Air pollution	Ozone concentrations	-0.4105*** (0.035)	-11.730
Density	Number of buildings	0.0353*** (0.006)	5.577
	Number of floors	-0.0738*** (0.021)	-3.455
	Number of residential units	-0.0299*** (0.007)	-4.323
	Number of residential and non-residential units	0.0287*** (0.007)	4.164
	The floor area ratio	0.2082*** (0.051)	4.109
	Residential floor area ratio	-0.2140*** (0.057)	-3.738
	Commercial floor area ratio	0.2357*** (0.047)	5.006
	Facility floor area ratio	0.1919*** (0.062)	3.072
Destination accessibility	Catering services [#]	0.9737*** (0.203)	4.794
	Shopping services [#]	-1.3866*** (0.217)	-6.396
	Residential services [#]	0.5157*** (0.169)	3.053
	Financial services [#]	2.1373*** (0.204)	10.502
	Life services [#]	-0.6163*** (0.177)	-3.474
	Recreational services [#]	-0.9410***	-6.085

Term		Estimate (std)	t
		(0.155)	
	Corporation services [#]	-0.3019*** (0.098)	-3.073
	Scientific services [#]	0.4813*** (0.106)	4.525
	Medical services [#]	0.5276*** (0.192)	2.752
	Government services [#]	-0.6365*** (0.125)	-5.088
Distance to transit	Transportation services	0.8775*** (0.199)	4.416
Design	Lot frontage	-0.0003*** (8.69e-05)	-3.241
	Building depth	0.0025*** (0.001)	2.177
	Building area	2.1655*** (0.337)	6.421
	Nodes	0.7783*** (0.076)	10.262
	Commercial area [#]	-1.4718*** (0.229)	-6.439
	Residential area [#]	-1.2069*** (0.416)	-2.900
	Retail area [#]	1.4660*** (0.200)	7.330
	Garage area [#]	-0.5863*** (0.190)	-3.090
Diversity	Mixed POI	5.7117*** (0.652)	8.758
Others	The number of housing unit	-0.0009*** (0.000)	-3.107
	Season	-0.4666*** (0.111)	-4.209
	Covid-19_(1: Yes; 2: No)	-1.1459*** (0.194)	-5.906
N		5679	
R ²		0.484	
R ² adjusts		0.481	

Note: ***: Significant at 99% confidence level. **: Significant at 95% confidence level

Variables transformed by z-score normalization

4.2. Model performance comparison

This study evaluates the performance of various machine learning models in predicting urban vitality before analyzing non-linear relationships, as presented in Table 3. This paper used

GBDT, RF and XGBoost models to compare the above OLS model. The dataset was split into training and testing sets at a ratio of 70:30, with 70% utilized for training/validation and the remaining 30% reserved as unseen data for testing. 5-fold cross-validation was employed during the training/validating process to identify the optimal parameters for predicting urban vitality to mitigate bias. The machine learning methods outperform the OLS model in urban vitality prediction. The performance differences among the machine learning algorithms are minor, with R^2 values ranging from 0.530 to 0.575 on the training dataset. XGBoost shows the lowest RMSE. XGBoost outperforms the other algorithms, achieving an R^2 of 0.560 and RMSE of 5.264, indicating a good performance on the testing dataset. Therefore, XGBoost was chosen as the highest accuracy algorithm for predicting urban vitality in 100x100m grid units of Manhattan, which aligns with previous studies on non-linear mechanism exploration. In the following steps, we use XGBoost to identify the variables' relative importance and non-linear relationships.

Table 3. Performance of the other three machine learning algorithms

Model	Training		Testing	
	RMSE	R^2	RMSE	R^2
GBDT	5.752	0.530	5.613	0.539
XGBoost	5.418	0.575	5.264	0.560
RF	5.648	0.547	5.596	0.542

4.3. Relative importance analysis

To identify the important features that impact urban vitality, this study ranked the relative importance of variables using SHAP. SHAP analysis reveals the extent to which the BE, road vehicles and air pollution contribute to urban vitality based on Shapley values. A factor with a higher Shapley value is more important in urban vitality prediction. The results of the SHAP analysis are presented in Figure 4 and Table 4. Fig. 4a illustrates how the feature values of each spatial unit contribute to urban vitality. Each data point represents a grid. For instance, the grid

provides higher nodes (red dot) that have higher SHAP values than those with lower nodes (blue dot). It means that the number of nodes positively influences urban vitality. In contrast, grids with higher ozone concentrations (red dot) have lower SHAP values than those with lower ozone concentrations.

Fig. 4b shows the relative importance of the BE, vehicles and pollution based on the Shapley values. Notably, four vehicle variables contribute 15.18 % to promoting urban vitality, with bicycles and buses accounting for 9.424%, further emphasizing the importance of public transport in enhancing urban vitality.

Regarding density dimension, the average commercial and facility floor area ratio contributed 4.921% and 1.257% to urban vitality, ranking 5th and 16th among the variables. Interestingly, the residential floor area ratio does not play a critical role in predicting urban vitality, as the SHAP value is minor. This result contradicts previous work on the linear relationship between residential density and urban vitality (Lv et al., 2022). A possible explanation is that increasing the floor area ratio may overemphasize residential buildings, neglecting other essential urban elements. As emphasized in prior studies, vibrant and lively urban areas necessitate a mix of residential, commercial, recreational, and cultural spaces (Akinci et al., 2022; Gómez-Varo et al., 2022).

Diversity and design are also significant contributions to urban vitality. As for the diversity dimension, the mixed POI contributes 4.607% of the influence on urban vitality. This result is supported by studies by Sung et al. (2013); Kang (2020); and Mouratidis and Poortinga (2020), who emphasized the importance of creating a diverse environment to attract locals and visitors to engage in activities on the street. Among design variables, the retail area is the most significant contributor, contributing 8.586% to urban vitality variance, followed by the number

of nodes and lot frontage, contributing 6.492% and 3.770% to urban vitality explanation in the Manhattan district, respectively.

The distance to transit and destination accessibility have significant contributions to urban vitality. Among the 15 most important variables, five are related to destination accessibility. In decreasing order, the variables related to destination accessibility are: financial services, catering services, medical services, shopping services, and scientific services. The total contribution of these variables is 31.52%. From Figure 4b, the most important feature for promoting urban vitality is financial services (such as banks, investment corporations and insurance), at 20.52%, followed by the retail area. Transportation services, contributing 3.66 % to urban vitality estimation, ranked 10th among the most important explanatory variables.

Ozone concentrations also have a significant negative impact on urban vitality. We observed that ozone concentrations contribute to 1.466% of urban vitality prediction. Compared with other built environment variables, the Shapley value of ozone concentration is significantly higher some built environment features; that is, the importance of ozone concentration is much higher than the 14 other built environment variables.

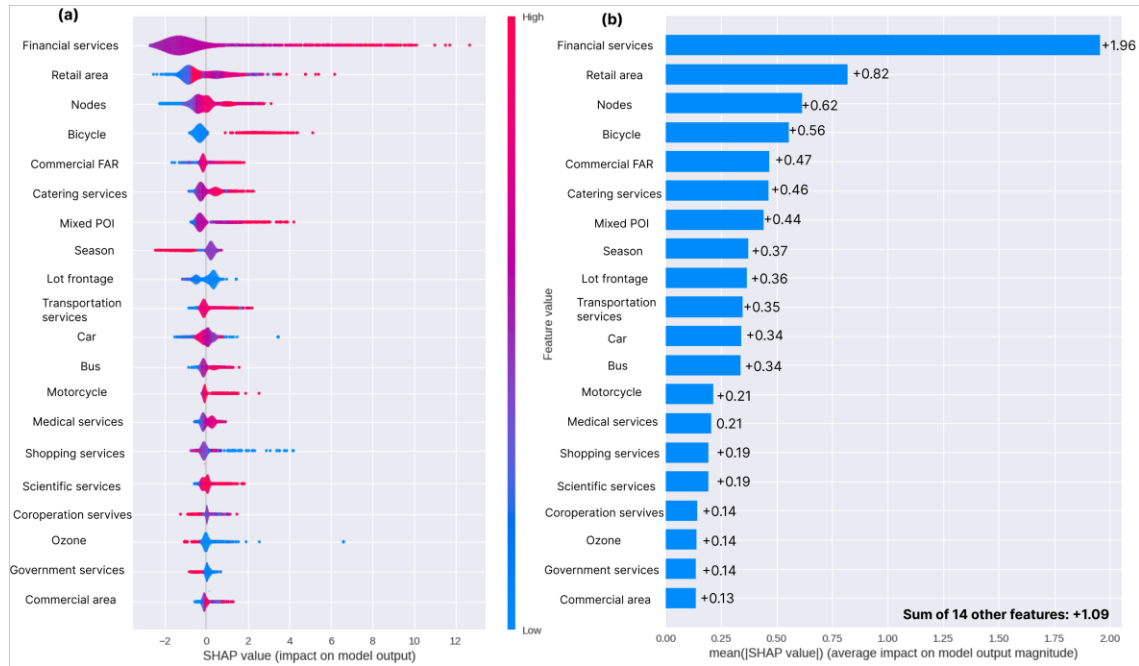


Figure 4. The relative importance of variables

Table 4. The contribution percentage and ranking of variables

Variables	Mean (SHAP value)	% Contribution	Important rank
Vehicles			
Bicycle	0.56	5.864	4
Car	0.34	3.560	11
Bus	0.34	3.560	11
Motorcycle	0.21	2.199	12
Pollution			
Ozone	0.14	1.466	14
Density			
The commercial floor area ratio	0.47	4.921	5
The facility floor area ratio	0.12	1.257	16
The number of buildings	0.11	1.152	17
Residential unit	0.1	1.047	18
Diversity			
Mixed POI	0.44	4.607	7
Design			
Retail area	0.82	8.586	2
Nodes	0.62	6.492	3
Lot frontage	0.36	3.770	9
Commercial area	0.13	1.361	15
Residential area	0.13	1.361	15
Distance to transit			
Transportation services	0.35	3.665	10

Variables	Mean (SHAP value)	% Contribution	Important rank
<i>Destination accessibility</i>			
Financial services	1.96	20.524	1
Catering services	0.46	4.817	6
Medical services	0.21	2.199	12
Shopping services	0.19	1.990	13
Science services	0.19	1.990	13
Corporation services	0.14	1.466	14
Government services	0.14	1.466	14
Life services	0.1	1.047	18
<i>Other variables</i>			
Season	0.37	3.874	8
Housing unit	0.12	1.257	16
<i>Sum of 8 other variables</i>	<i>0.43</i>	<i>4.503</i>	<i>-</i>
Sum	9.55	100%	-

4.4. Non-linear effects of road vehicles, built environment on urban vitality

This section plots the nonlinear interaction relationships of 23 critical variables on urban vitality. These variables can explain the 91.4% of urban vitality variance.

4.4.1. Non-linear and threshold effects of road vehicles, pollution on urban vitality

We observe a positive non-linearity between the appearance of vehicles and urban vitality (figure 5). The number of bicycles on the street exhibits a threshold influence on urban vitality (figure 5a). Beyond 20 cars, we observe a positive association with urban vitality, accompanied by a slight increase in the corresponding Shap value as the number of cars rises (figure 5b). Similarly, as shown in Figure 5c, the presence of buses is associated with the promotion of urban vitality. The influence of buses on urban vitality is highest when the number of buses is in the range of 4 to 6. This finding aligns with studies on the association between public transport and walking (Wasfi et al., 2013; Beavis and Moodie, 2014). Notably, a high number of buses, combined with enhanced shopping accessibility, correlates with increased urban vitality. This underscores the significance of strategically planning transportation routes and optimizing urban land use to bolster human activity. The number of motorcycles also enhances urban vitality

(figure 5d). The presence of motorcycles also shows a positive correlation with improved urban vitality. These findings align with prior qualitative studies, emphasizing the positive correlation between bicycle and bike-sharing volumes and urban vitality. It is noteworthy that many previous quantitative studies often overlook the impact of these vehicles on urban vitality.

We also noted a negative correlation between ozone concentrations and urban vitality. There is a threshold at an ozone concentration of 27.5ppb, at which the influence on urban vitality remains stable.

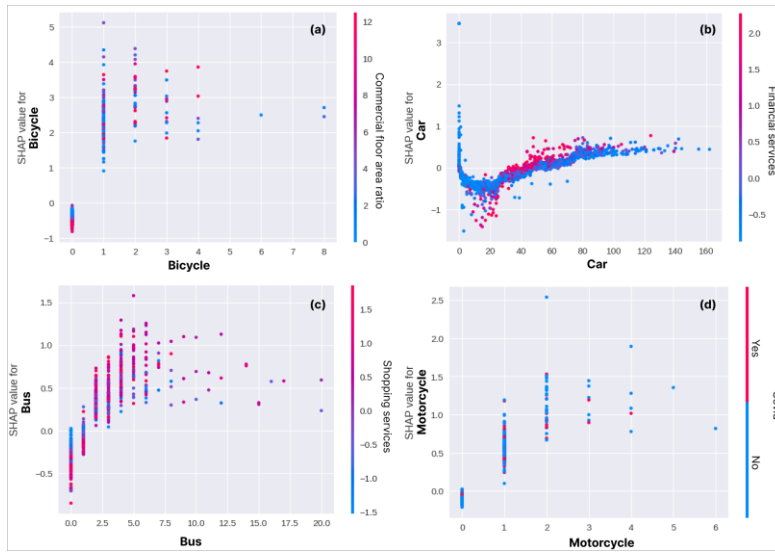


Figure 5. Non-linear relationship between vehicles and urban vitality

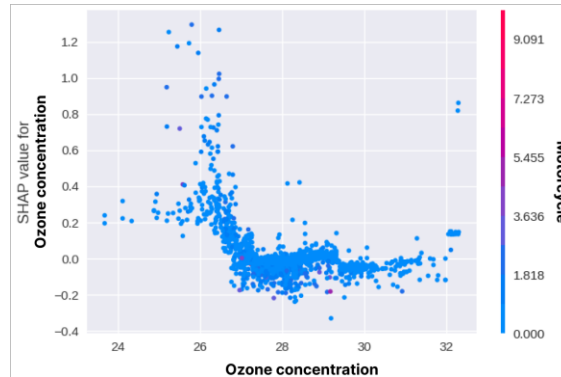


Figure 6. Non-linear relationship between ozone concentrations and urban vitality

4.4.2. Non-linear and threshold effects of the built environment on urban vitality

4.4.2.1. Urban vitality and diversity and distance to transit

We observed a positive influence of distance to transit and diversity features on urban vitality. Notably, a threshold was identified at a transportation service score of 1, beyond which the Shap value significantly increased, indicating a higher influence on urban vitality (Figure 7a). The synergy effect of transportation accessibility and floor area ratio contributed to a higher Shap value, signifying increased urban vitality. This observation underscores the pivotal role of enhanced transportation planning, as higher transportation accessibility and floor area ratio will encourage walking activities, indicative of elevated urban vitality levels. Likewise, the Shapley values of the mixed POI showed a non-linear relationship (figure 7b). Specifically, the influence of the mixed POI on urban vitality is minimal when the POI mix value ranges from 0.1 to 0.4. However, once the mixed POI surpasses the critical threshold of 0.4, the impact on urban vitality experiences a rapid upswing. Interestingly, we observed that areas with high mixed Points of Interest (POI) values experienced reduced pedestrian volume during COVID-19 compared to non-COVID periods. This finding aligns with the Protection Motivation Theory of Boer and Seydel (1996), as during the COVID-19 pandemic, pedestrians may tend to avoid areas with a high concentration of mixed POIs to mitigate the risk of infection.

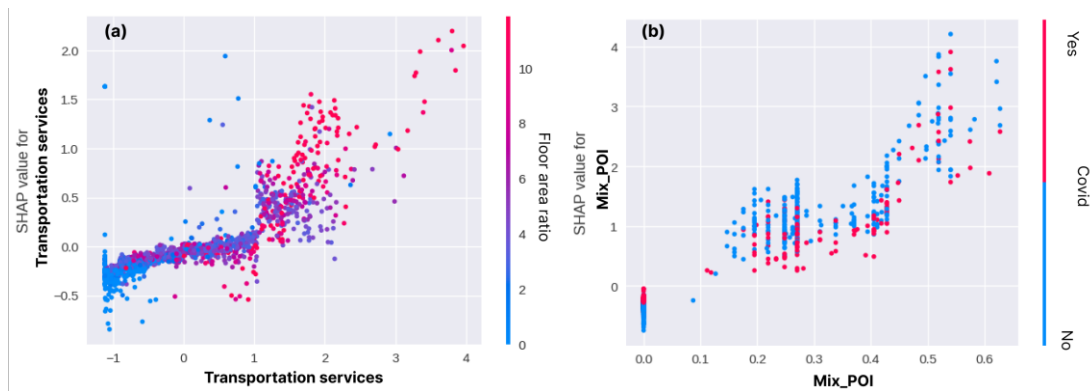


Figure 7. Non-linear relationship between transportation, mixed POI and urban vitality

4.4.2.2. Urban vitality and density features

Figure 8 illustrates the non-linear relationship between density dimension and urban vitality. In general, the density dimension positively affects urban vitality. The commercial floor area ratio (CommFAR) emerged as a significant factor in promoting urban vitality. The Shap value slightly increases as CommFAR is in the range of 0-6. With subsequent growth in CommFAR, there is a more pronounced influence on urban vitality prediction (figure 8a). A similar pattern is observed in the facility floor area ratio, although its effect is minor (Figure 8b). It was also observed that the effect of the number of buildings remains constant after a threshold of 15, indicating that an extremely high building density may not be further associated with urban vitality enhancement (figure 8c).

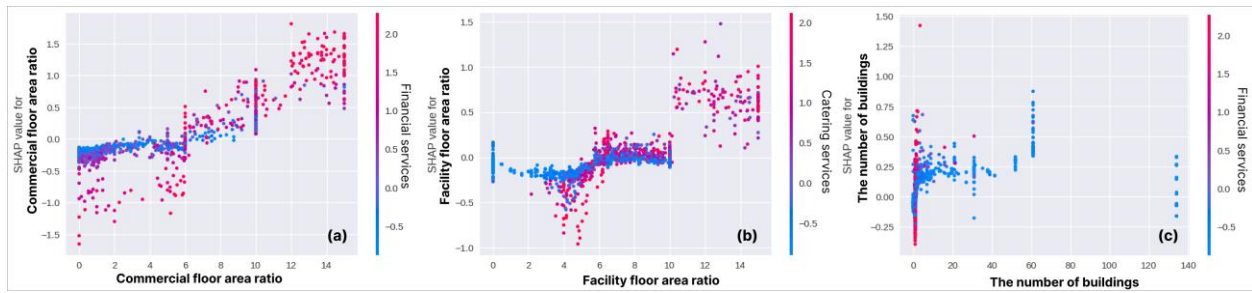


Figure 8. Non-linear relationship between density dimension and urban vitality

4.4.2.3. Urban vitality and design features

Figure 9 illustrates non-linear relationships between design features and urban vitality. Area use for commercial and retail purposes also positively influences urban vitality (see Figures 9a and 9c). Nodes significantly enhance urban vitality, as indicated by Shap values. However, the Shap values slightly decrease when the number of nodes in a 100m grid exceeds 5. This finding underscores the importance of intersection presence in promoting vitality. An excessively high density of nodes might not further improve urban vitality (figure 9b). This finding adds nuance to previous studies (Chen et al., 2021b; Chen et al., 2022b) that reported mixed results regarding the impact of road intersections on urban vitality by revealing non-linear mechanisms. In

contrast, the average lot front exhibits a negative non-linear association with urban vitality. The SHAP value dramatically decreases as the lot front increases to 1000 feet. Beyond this threshold, the influence on urban vitality remains stable (Figure 9d). This finding aligns with previous studies (Gan et al., 2021), emphasizing the significance of small blocks in promoting urban vitality. Figure 9e illustrates a non-linear relationship between urban vitality and residential areas in a grid. The optimal score for residential areas to promote urban vitality is 0, indicating appropriate distribution for different purposes. As the residential area score exceeds 0, the influence on urban vitality becomes negative.

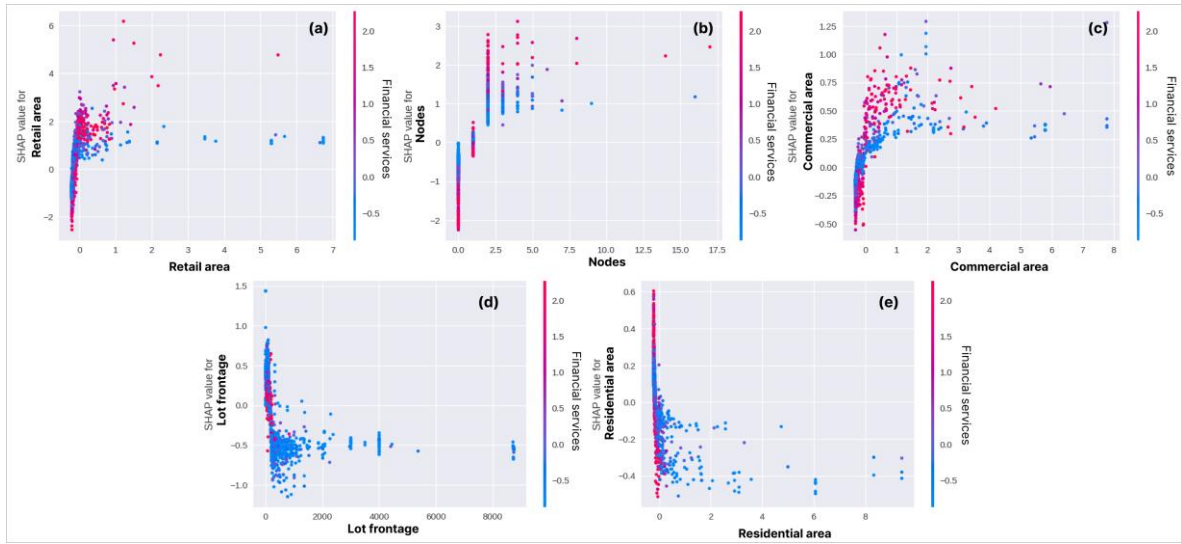


Figure 9. Non-linear relationship between design dimension and urban vitality

4.4.2.4. Urban vitality and destination accessibility features

As depicted in Figure 10, destination accessibility variables exhibit non-linearity correlated with urban vitality. Financial services are crucial in enhancing urban vitality (figure 10a). We observed a synergistic impact between financial and retail services on urban vitality, as indicated by higher SHAP values corresponding to increased levels of these services. Catering services also positively affected urban vitality, as depicted in Figure 10b. The local effects transitioned from negative to positive when the catering service score surpassed 0. However, the

promoting impact diminished once the catering service score exceeded a threshold of 2. Additionally, the relationship between medical services and urban vitality followed an inverted U-shaped trend, peaking at 0.9 (Figure 10c). Medical services exhibited a positive correlation with urban vitality when the score was below 0.9, but led to a decline in urban vitality if the score exceeded this threshold. Figure 10d demonstrated a clear nonlinear association between shopping services and urban vitality. Moreover, scientific services showed a positive correlation with urban vitality, with the local effect transitioning from negative to positive when the scientific service score exceeded 2 (Figure 10e). In contrast, increased government services were associated with decreased urban vitality once the threshold of 2 was surpassed, after which the local effect stabilized (Figure 10f).

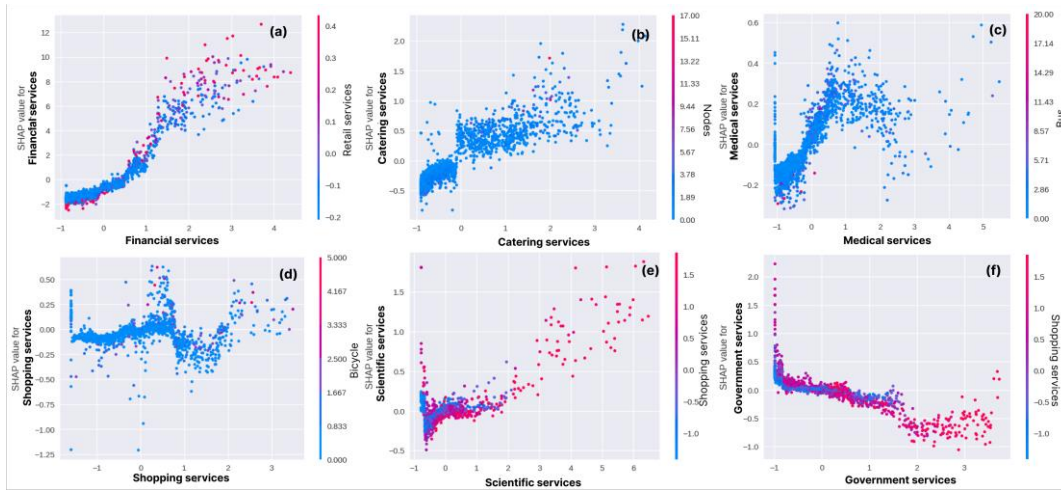


Figure 10. Non-linear relationship between destination accessibility and urban vitality

4.4.3. Impacts of season and covid on urban vitality

Our findings reveal that urban vitality is lowest during the winter months (Figure 11a), which is likely influenced by low temperatures and the COVID-19 outbreak in New York City (NYC) during the winter of 2021. As documented by Hunter et al. (2021), people restricted both essential and non-essential travel during the pandemic. They adjusted their movement patterns in response to the epidemic, which aligns with the Protection Motivation Theory (Boer and Seydel,

1996). With a surge in COVID-19 cases, perceived health risks increased, reducing individuals' inclination to engage in street activities. Additionally, stringent governmental policies to curb mobility further contributed to decreased urban vitality.

Urban vitality showed improvement during the spring, as evidenced by increasing Shapley values. Similarly, during the summer and autumn months, SHAP values rose, indicating a positive influence on urban vitality. Areas with a greater presence of catering services exhibited higher Shapley values during summer and autumn, reflecting heightened vitality levels. The implementation of post-COVID reopening policies and vaccination strategies by the NYC government played a pivotal role in restoring urban vitality in the Manhattan district (Li and Lasenby, 2023) in summer and autumn. The significant decline in the average daily number of confirmed and probable positive COVID-19 cases during the summer of 2021, is consistent with the findings of Xi et al. (2024), underscoring the positive impact of vaccinations in revitalizing travel behaviors.

Furthermore, we observed a synergy effect between COVID-19 and mixed POI. During the COVID-19 pandemic, areas with lower mixed POI values experienced higher pedestrian volumes, indicating heightened urban vitality. Conversely, as the COVID-19 wave subsided, places boasting diverse urban amenities attracted more pedestrians.

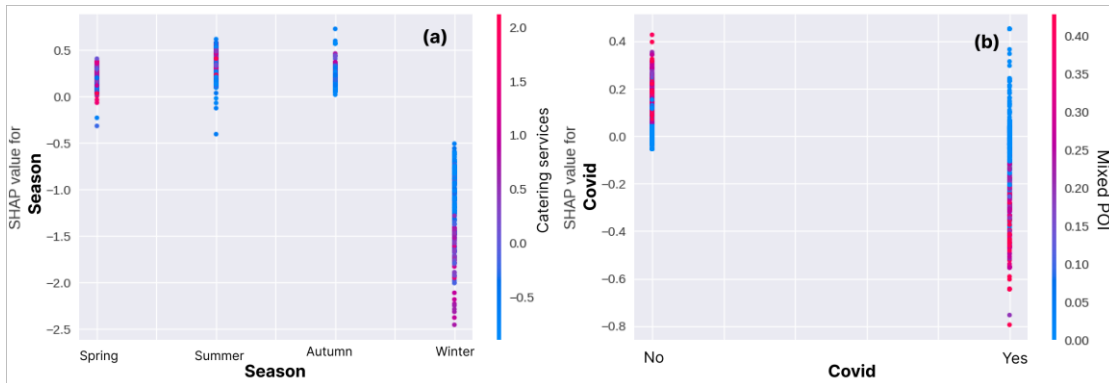


Figure 11. Non-linear relationship between season, COVID-19 and urban vitality

4.4.4. Interaction effects between variables

We have introduced an interaction chart featuring the top 10 variables to provide a more comprehensive understanding of the interaction effects among the built environment, road vehicles, and air quality (Figure 12). This visual representation offers an overview of how these variables influence urban vitality. In generating the interaction chart, we computed the absolute Shapley values of each variable concerning others, thereby quantifying their interaction effects. Each graph in the chart represents a unique pair of variables, showing how changes in one variable affect urban vitality as the other variable varies. The X-axis denotes the selected variable, while the color axis on the right illustrates the high and low values of the other variable, with red denoting high values and blue indicating low values. Furthermore, the Y-axis depicts their combined influence on urban vitality. Negative values below 0 signify antagonistic effects (i.e., negative impacts), whereas positive values above 0 indicate a positive impact.

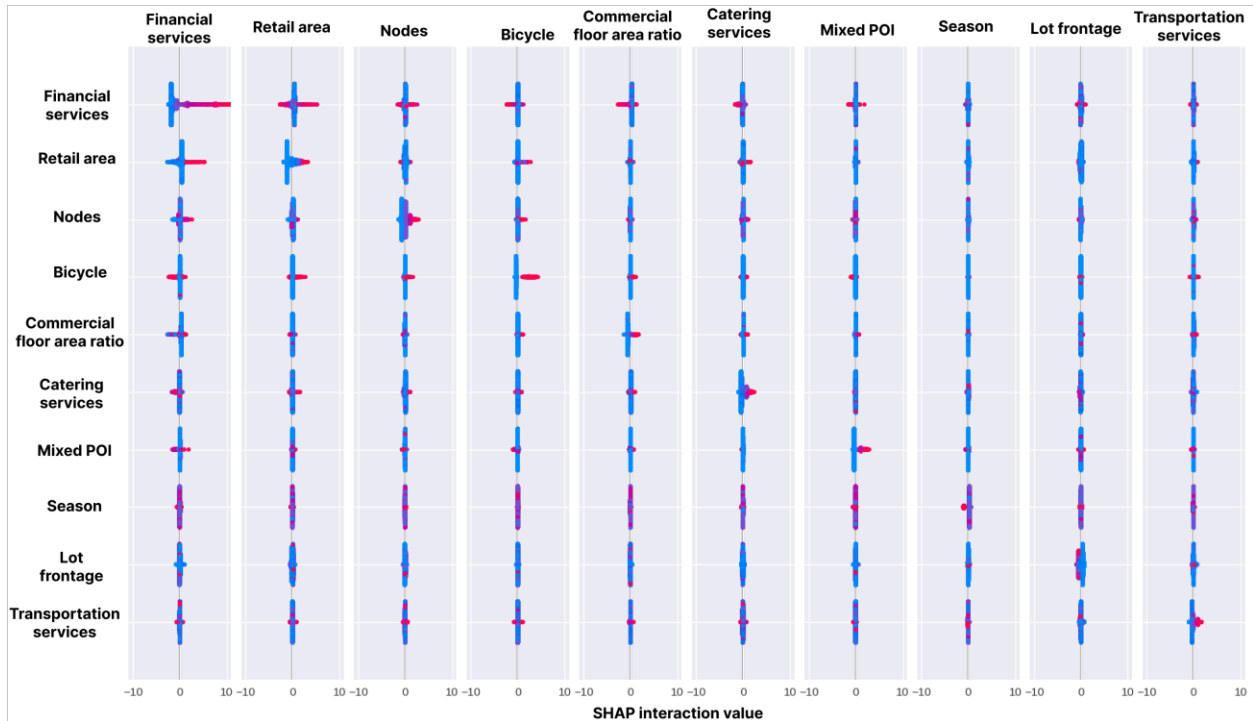


Figure 12. SHAP interaction value of 10 key variables

4.5. Local and spatial effects on urban vitality

To elucidate the spatial disparities in the influences of the built environment, pollution, and road vehicles, we employed a clustering approach to classify grids based on similar patterns of local effects. The optimal number of clusters was determined using the Silhouette Coefficient and the Davies-Bouldin indexes. Figure 13 visually represents the local explanations for four selected grids within the Manhattan district, each corresponding to distinct clusters. These selected grids showcase varying levels of urban vitality across the study area, offering insights into how specific feature values contribute to predicting urban vitality for each grid.

In Cluster 1, there is a low urban vitality prediction. For example, in grid number two, urban vitality is negatively affected by financial services, nodes, lot frontage, bicycles, mixed POI, and catering services (indicated by blue). As a result, the urban vitality prediction value for this grid is low. The low vitality value of case number two can be attributed to its proximity to Central Park, where the built environment is less pronounced.

Cluster 2 also exhibits relatively low urban vitality prediction. For instance, in the first grid, key influencing factors include a retail area, motorcycle, lot frontage, and ozone (indicated by red). Conversely, financial services, bicycles, buses, and mixed POI contribute to lower urban vitality values.

Cluster 3 has a medium predicted urban vitality. For example, the fifth case, located in Manhattan's financial district, demonstrates a markedly higher vitality level. This area is characterized by a high concentration of financial services, nodes, diverse POI, retail, and commercial services.

Cluster 4 exhibits the highest predicted urban vitality. For instance, grid three, situated at the centre of socio-economic activities, exhibits a higher urban vitality prediction. Factors such

as financial services, retail areas, floor area ratio, buses, POI mix, floor area ratio, and transportation services prominently contribute to urban vitality (indicated by red).

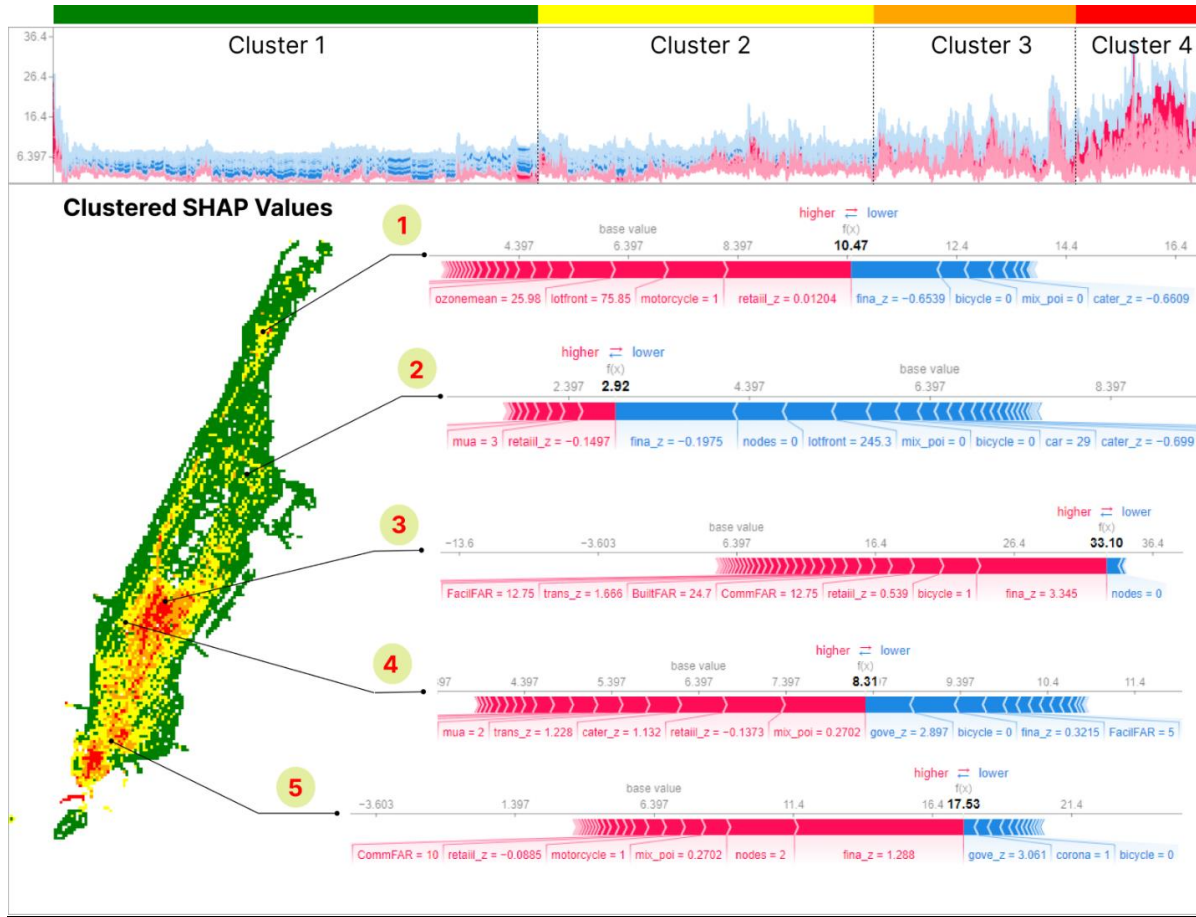


Figure 13. Clustered SHAP value and local effects by cluster type

5. Discussion

5.1. Non-linear relationships between urban vitality and the built environment

Our findings in part concur with existing literature, exhibiting the threshold effects of the built environment on urban vitality. Specifically, the diversity of services plays a crucial factor in enhancing the liveliness of an area, consistent with various previous studies (Mouratidis and Poortinga, 2020; Li et al., 2021a). However, the impact of the mixed POI on urban vitality is modest, with mixed POI values ranging from 0.15 to 0.4. When the value surpasses 0.4, the influence on urban vitality significantly intensifies. This aligns with human preferences, as people tend to be attracted to areas with diverse services and amenities (Chen et al., 2021a).

Monofunctional areas, in contrast, tend to exhibit a lower vibrancy level. Similarly, the impact of transportation services on urban vitality becomes more pronounced when the transportation service score exceeds 1.0, particularly in conjunction with a high facility floor area ratio. This underlines the importance of coordinating transportation and urban facility planning to foster a vibrant city, as a substantial increase in SHAP values is indicated in Figure 7a. A deficiency in urban facilities and limited transportation accessibility can negatively impact people's engagement in non-work activities (Badland and Schofield, 2005; Farber et al., 2018).

Densities, particularly facility floor area ratio and commercial floor area ratio, significantly contribute to urban vitality, which aligns with previous research (Ye et al., 2018; Lu et al., 2019; Liu and Shi, 2022). However, we noted a negative influence when the built floor area ratio reaches its maximum value, indicating that excessive density may not enhance urban vitality. This finding contradicts linear-based studies, which reported a positive impact between the built floor area ratio and urban vitality (Yang et al., 2021). While an increase in the residential floor area ratio implies a rise in population, our observations suggest that solely focusing on this aspect is insufficient for fostering urban vitality. It requires complementary conditions, such as providing essential urban services like catering, life, and financial services. Human mobility tends to encourage the development of more open spaces and the provision of catering services, which aligns with findings from prior studies (Koohsari et al., 2015; Zhong et al., 2020).

From the design perspective, the results indicate that certain design features exhibit varying levels of influence on urban vitality. Notably, the number of nodes significantly enhances urban vitality, aligning with studies by Long and Huang (2019); Chen et al. (2021b). However, this finding contradicts studies by Li et al. (2022b) and Chen et al. (2022b), which reported a

negative influence of street intersections on urban vitality. These mixed results could be attributed to a lack of consideration for non-linearity and interaction effects among variables. Using XGBoost and SHAP, we observed a threshold effect of node density on urban vitality. Specifically, as the density of nodes increases up to a threshold of 5, there is a positive association with urban vitality. However, beyond this threshold, further increases in node density are no longer correlated with a significant enhancement in urban vitality. This threshold effect suggests that a certain level of node density is necessary to promote urban vitality. Excessively high densities may not yield additional benefits and could have diminishing returns, such as traffic congestion and air pollution (Xie et al., 2019). Our findings underscore the significance of small blocks in enhancing urban vitality, consistent with the ideas presented by Jacobs (1961) and Li et al. (2020). Also, we found that the optimal lot depth for maintaining urban vitality is about 200 to 400 feet. Beyond this range, the influence is relatively minor. This result supports the study of Gan et al. (2021), who hypothesized a non-linear effect between urban vitality and block size. In urban planning, walkability and connectivity are paramount for sustaining urban vitality. Lots with depths within the 200 to 400-feet range tend to promote walkability, making it easier for residents to access amenities, public spaces, and transportation nodes (Sevtsuk et al., 2016). Beyond this range, the increased distance between destinations may lead to decreased walkability and connectivity, diminishing the area's vitality.

This study looks further into how destination accessibility influences urban vitality in different ranges. For example, the high density of financial services accounts for a fifth in explaining urban vitality, while an excessive concentration of medical services is associated with a less vibrant environment. This aligns with expectations, as Manhattan's status as an international financial center of NYC naturally leads to financial services playing a primary role

in enhancing urban vitality. These services stimulate local businesses, attracting more residents and workers to the area (Wang et al., 2022a). Consequently, the community becomes vibrant and dynamic, hosting various commercial and social activities. Planners and policymakers can strategically design neighborhoods that integrate housing with financial centers, creating walkable and accessible areas that foster social interaction and economic vitality based on this result. Additionally, an excessive concentration of medical services (beyond a certain threshold) can indicate a low-vitality environment. This intriguing finding suggests that an overabundance of medical services may be linked to health issues or high disease rates, potentially influencing the overall vibrancy and liveliness of the place. However, this finding should be further explored at larger scales, such as district and city levels. Furthermore, the results indicated that the combination of residential and financial services significantly contributed to nurturing a lively environment. Catering services, including restaurants, cafes, and food stalls, provide spaces for people to gather and interact (Yan, 2013). These places serve as social hubs where residents and visitors can meet, socialize, and converse. Such social spaces foster a sense of community and contribute to the overall liveliness of the urban environment. These findings supported the results in studies by Ye et al. (2018); Zhang et al. (2020a); Fu et al. (2021); and Wu et al. (2022b).

5.2. The threshold effects of road vehicles and air pollution on urban vitality

This work confirmed that a vibrant environment is closely tied to the presence of vehicles. The results emphasize the significance of bicycles and buses in engaging people in street activities. Among various modes of transportation, we found that bikes make the most substantial contribution, followed by cars and buses. It is worth noting that the presence of road vehicles is often overlooked in urban vitality measurement and its determinant studies (Zeng et al., 2020). Furthermore, we identified synthetic effects between vehicles and financial and shopping services. This finding suggests that urban planners and policymakers can promote

lively neighborhoods by considering initiatives such as bike-sharing programs and designing optimal bus routes near financial and shopping services. While the presence of cars shows a positive association with urban vitality, urban management should be mindful of the dual impact of private vehicles. While private cars offer convenience and travel flexibility, they can also contribute to congestion and pollution (Yang, 2023), potentially negatively influencing urban vitality.

We also observed the negative threshold effect between ozone concentration and urban vitality (Figure 6). Ozone concentration related to urban vitality decreases within the range from the minimum to near average value (23.67 ppb to 27.5 ppb). Elevated levels of air pollution can discourage people from engaging in outdoor activities such as walking, cycling, and socializing in public spaces (Jiang et al., 2019). This can lead to declining social interactions and community engagement, which are essential for a vibrant urban environment. Additionally, individuals may be more willing to pay to avoid areas with poor air quality (Enrico and Matthew, 2011) or even consider moving to cleaner air (Pan, 2023). This study examined several pollutants, including PM_{2.5} and O₃; however, only the O₃ concentration showed statistical significance. This finding is consistent with observations in NYC, where PM_{2.5} levels have decreased by 43% since 2009, while O₃ concentrations remain persistently high. According to the American Lung Association, which assigns grades to counties based on air quality, New York is ranked 12th out of 227 cities for days with high ozone. Out of our expectation, urban vitality remains unchanged when ozone concentration surpasses the threshold of 27.5 ppb. We hypothesize that there is a trades-off between air pollution exposure and participation in street activities. Although people are aware of the negative impact of pollution on their health, they still participate in street activities. This is understandable given that the contribution of ozone concentration to urban vitality is minor, at

1.46%. However, we believe measures should be implemented to minimize the negative impacts of air pollution on human health and activities, thereby maintaining a vibrant city.

5.3. Planning and policy implication

Our findings offer several implications for urban planning and policymaking. Firstly, enhancing urban vitality necessitates the provision of not only efficient transportation services but also a variety of urban amenities. Our analysis reveals notable interaction effects among transportation, financial, residential, and catering services, underscoring the importance of accessibility to foster urban vitality. Planners could refer to dependent plots of transportation and urban amenities to optimize the density and diversity of these features, thereby facilitating the attainment of desired levels of urban vitality. Secondly, our study underscores that urban vitality is influenced not only by built environment factors such as the density of financial, transportation, and catering services but also by air quality and vehicular traffic flows. This aligns with the three-factors theory (basic, performance and excitement factors) (Kano et al., 1984). Basic elements, such as built environment features, are necessary but insufficient for urban vitality. Focusing on performance and excitement factors will further promote urban vitality. For example, progressive reductions in pollution levels have a proportional impact on urban vitality, but diminishing returns might be observed. Initiatives for a cleaner environment, such as green transportation, can create significant non-linear improvements in urban vitality. Decision-makers should consider the interaction effects of these diverse features when formulating urban planning strategies. Understanding these interactions empowers urban planners and policymakers to make more informed decisions regarding city design and planning. Thirdly, the built environment features exhibit nonlinear associations with urban vitality, with variable-specific upper and lower thresholds. Failure to account for nonlinearity may lead to underestimation of the impacts of certain variables. It is recommended that planners identify

critical values built environment features must attain to enhance urban vitality effectively. For instance, maintaining a mixed Point of Interest (POI) ratio of around 0.52 ensures that POI diversity optimally contributes to urban vitality promotion. Lastly, expanding the scope of urban vitality assessment beyond global parameters to encompass detailed aspects of road vehicle pollution control is advisable. Given that urban vitality is influenced by various factors, including the built environment, road vehicles, and pollution, implementing the above implications necessitates the collaborative efforts of multiple stakeholders, including urban planners, environmentalists, and transportation managers.

6. Conclusion

This study took Manhattan as the study area to explore the non-linear relationship between urban vitality and the BE, road vehicles and air pollution. Our findings indicate that a good built environment is essential for fostering urban vitality, accounting for 77.5% of urban vitality explanations. Sufficient financial services, retail areas, diverse points of interest, and intersection nodes emerged as the most significant contributors to urban vitality. Additionally, vehicles such as bicycles, buses, and motorbikes contribute 15.18% to urban vitality estimation, while ozone pollution contributes 1.46%. We observed the non-linear effects of BE features on urban vitality; that is, the built environment features may reduce urban vitality in specific ranges. An inverted U-shaped relationship is observed between urban vitality and medical services. While road vehicles exhibit positive threshold effects on urban vitality, air pollution has a negative threshold impact. Through our findings, we discussed the implications of non-linear relationships for urban planners and policymakers to foster urban vitality. This study contributes to the literature by validating the urban vitality theory and exploring synergies and non-linearity threshold effects between urban vitality and BE, road vehicles and air pollution.

While this study provides new conceptual contributions to urban sustainable development, it also has some limitations that should be acknowledged. Firstly, although the street view images used in this study can reveal pedestrian flows and vehicle volumes based on the spatial dimension, capturing the temporal units (morning/afternoon) remains challenging. Additionally, we did not explore the potential differences in pedestrian flow during different periods, such as peak/non-peak hours. This issue is a common limitation of street view image data (Li et al., 2022b; Yang et al., 2024). Secondly, the proxy of urban vitality is extracted from street view images that may not account for significant indoor activities. Future studies can consider integrating additional data sources, such as location-based services and social check-in data, to capture temporal variations in urban vitality. Finally, while machine learning algorithms outperform conventional models in prediction performance and non-linearity analysis, they do not establish causation between variables. Future studies should conduct longitudinal designs to explore causal relationships.

Appendix 1: POI categories

Type	Categories
Catering services	Restaurant, pub, coffee, dessert, etc
Shopping services	Shopping malls, department stores, markets, supermarkets, etc
Residential services	Hotel, inn, apartment
Financial services	Bank, investment corporation, insurance, company, etc
Life services	Barbershop, dry cleaner, telecom office, etc
Recreational services	Cinemas, KTV, Pool room, game room, etc
Corporation services	Technology, design and consulting institution
Science, technology, school	School, University, Academy, Museum, Gymnasium, etc
Medical services	Hospital, Clinic, Drugstore, etc
Transportation services	Railway and subway station, bus station, bike-sharing station
Government and social organization	Governmental Agencies, neighbourhood committee

Appendix 2: Data descriptive analysis

Term	Variables	Definition	Min	Max	Mean	S.tD
Urban vitality	Pedestrian	The number of pedestrians captured by street view	0	102	6.403	8.310

Term	Variables	Definition	Min	Max	Mean	S.tD
Vehicle		images				
	Car	The number of cars detected in street view images	0	221	45.89	28.501
	Bicycle	The number of bicycles detected in street view images	0.172	9.0	0.172	0.576
	Bus	The number of buses detected in street view images	0	15	0.895	1.652
	Motorcycle	The number of motorcycles detected in street view images	0	22	0.181	0.603
Pollution	Ozone	The summer average Ozone concentration (ppb)	23.012	32.46	28.072	1.204
	PM _{2.5}	Annual average fine particulate matter <2.5 microns (µg/m ³)	6.193	9.687	7.345	0.801
	NO ₂	Annual average Nitric dioxide (ppb)	12.46	27.54	19.686	2.841
Density	NumBldgs	The number of buildings	0	134	6.866	19.293
	NumFloors	The number of floors	0	104	7.809	7.653
	BuiltFAR	The built floor area ratio	0	29.763	4.276	3.782
	ResidFAR	The residential floor area ratio	0	10	4.469	3.022
	CommFAR	The commercial floor area ratio	0	15	2.335	3.948
	FacilFAR	The facility floor area ratio	0	15	5.788	3.446
	Unitres	The number of residential units	0	8764	167.15	825.67
	Unitatotal	The number of residential non-residential units	0	8812	173.817	833.39
Diversity	POI mix	POI diversity	0	0.769	0.074	0.147
Design	Edges	The sum of the total length of edges within the cell (m)	0	1449.16	174.53	108.59
	Nodes	Number of intersections	0	20	0.826	1.153
	Lotfront	The lot's frontage (feet)	0	8744.2	377.01	1026.50
	Lotdepth	The lot's depth (feet)	0	5000	312.387	712.350
	Bldgfront	The building's frontage along the street (feet)	0	1362	74.270	100.970
	Bldgdepth	The lot's depth (feet)	0	1226.5	94.11	100.34
	Unittotal	The number of residential and non-residential units	0	8812.0	172.83	831.14
	Building -z	The total gross area, except condominium (square feet) converted to z-score	-0.345	10.868	0	1.0
	Commer-z	Area used for commercial purposes (square feet)	-0.323	9.444	0	1.0

Term	Variables	Definition	Min	Max	Mean	S.tD
		converted to z-score				
	Resarea-z	Area used for residential purposes (square feet) converted to z-score.	-0.213	9.417	0	1.0
	Retail-z	Area used for retail purposes (square feet) converted to z-score	-0.208	9.207	0	1.0
	Garage-z	Area used for garage purposes (square feet) converted to z-score	-0.196	10.754	0	1.0
	Officearea-z	Area used for office purposes (square feet) converted to z-score	-0.246	14.3	0	1.0
	Factryarea-z	Area used for factory purposes (square feet) converted to z-score	-0.045	38.5	0	1.0
	Strgarea-z	Area used for storage/loft purposes (square feet) converted to z-score	-0.067	37.2	0	1.0
	Otherarea-z	Area used for other purposes (square feet) converted to z-score	-0.184	19.5	0	1.0
Distance to transit	Transport-z	The density of railway and subway stations, bus stations, bike-sharing stations converted to z-score	-1.113	3.965	0	1.0
Destination accessibility	Catering-z	The density of catering services converted to z-score	-0.896	4.398	0	1.0
	Shop_z	The density of shopping services converted to z-score	-1.593	3.629	0	1.0
	Resid-z	The density of residential services converted to z-score	-0.6388	5.036	0	1.0
	Financial z	The density of financial services converted to z-score	-0.870	4.582	0	1.0
	Life-z	The density of life services converted to z-score	-0.827	3.896	0	1.0
	Recreation-z	The density of recreational services converted to z-score	-0.615	4.494	0	1.0
	Corporation-z	The density of corporational services converted to z-score	-0.400	12.78	0	1.0
	Science-z	The density of science services converted to z-	-0.765	6.568	0	1.0

Term	Variables	Definition	Min	Max	Mean	S.tD
Others		score				
	Medicine-z	The density of medical services converted to z-score	-1.055	5.480	0	1.0
	Government-z	The density of governmental services converted to z-score	-0.982	3.733	0	1.0
	Population density	The population density	0.103	2579.077	177.570	192.818
	Season (mua)	Season in year (Spring: 1; Summer: 2; Autumn: 3; Winther: 4)	1.0	4.0	2.428	0.879
	Covid	(Yes: 1; No:0)	0	1.0	0.306	0.460
	Housing unit	The number of housing units	276.224	331.512	0	3479.0

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